

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing

Techniques for analyzing Computer Vision systems. (BTL 3)

Assessment Tool Weightage: 40%

Code of Conduct:

I **K Sampath Reddy (192424285)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K Sampath Reddy

192424285

Question 3

Image acquisition, sampling, quantization, resolution and sensing are fundamental concepts in computer vision. In this scenario, the correct design of the imaging system directly influences the quality of the captured image and the accuracy of later processing stages. When the image is captured with appropriate spatial sampling and sufficient intensity resolution, important visual details are preserved while unwanted artifacts are minimized. If sampling is too low, information is permanently lost and aliasing or blurred edges may appear. Likewise, insufficient quantization reduces the number of intensity levels, producing banding and loss of subtle details. Modern computer vision systems overcome these issues through proper camera calibration, suitable sensors, controlled lighting, preprocessing techniques such as filtering, contrast enhancement, and interpolation, followed by robust feature extraction algorithms. The final result is improved detection, recognition, measurement, and decision making in applications such as medical imaging, surveillance, autonomous vehicles, OCR, satellite imaging, and industrial inspection. Therefore, selecting appropriate acquisition parameters and image processing techniques is essential for achieving reliable and accurate computer vision performance.

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